

Lab Project:
**DoS & DDoS Attack Simulation and
Traffic Analysis**

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1. Lab Task Objectives / Description:

The objective of this lab is to create two VM's so that an isolated environment is created then use hping3, tcpdump and Wireshark to analyze patterns of DoS and DDoS.

2. Software's / Tools Required:

- Hardware: PC, Internet access
- Software: Wireshark, tcpdump, Hping3, VMWare
- VM's: Ubuntu (target-Host Only, NAT), Kali(attacker-Host only)

3. Lab Procedure & Results:

PART 1: Basic Ping and Port Scanning

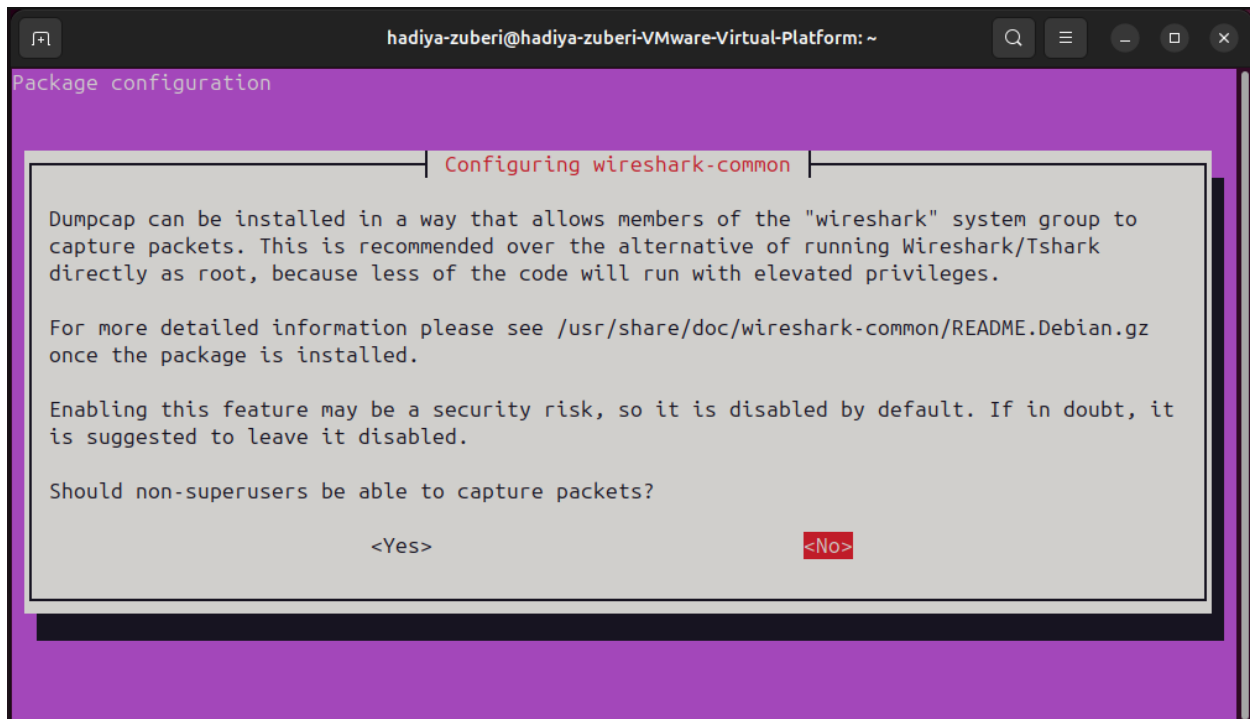
STEPS:

1. Install Tcpdump and wireshark on target machine (Ubuntu) using commands

sudo apt update

sudo apt install wireshark tcpdump -y

Type "yes" when asked if non superusers are allowed to capture packets



2. Add yourself to wireshark group using: `sudo usermod -aG wireshark $USER`

Then restart Ubuntu

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform: ~  
Setting up libegl-mesa0:amd64 (25.2.8-0ubuntu0.24.04.1) ...  
Setting up libswresample4:amd64 (7:6.1.1-3ubuntu5) ...  
Setting up libqt6quick6:amd64 (6.4.2+dfsg-4build3) ...  
Setting up libavcodec60:amd64 (7:6.1.1-3ubuntu5) ...  
Setting up mesa-va-drivers:amd64 (25.2.8-0ubuntu0.24.04.1) ...  
Setting up libswscale7:amd64 (7:6.1.1-3ubuntu5) ...  
Setting up libglx-mesa0:amd64 (25.2.8-0ubuntu0.24.04.1) ...  
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...  
Setting up libqt6waylandeglcompositorhwinintegration6:amd64 (6.4.2-5build3) ...  
Setting up libavformat60:amd64 (7:6.1.1-3ubuntu5) ...  
Setting up libqt6waylandeglclienthwinintegration6:amd64 (6.4.2-5build3) ...  
Setting up va-driver-all:amd64 (2.20.0-2ubuntu0.1) ...  
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...  
Setting up libqt6multimedia6:amd64 (6.4.2-11build3) ...  
Setting up ubuntu 24.04.1 LTS amd64 (4.2.2-1.1build3) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.3) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for shared-mime-info (2.4-4) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
Processing triggers for hicolor-icon-theme (0.17-2) ...  
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...  
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ sudo usermod -aG wireshark $USER  
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$
```

4. On Kali (Attacker machine) install hping3 using:

`sudo apt update`

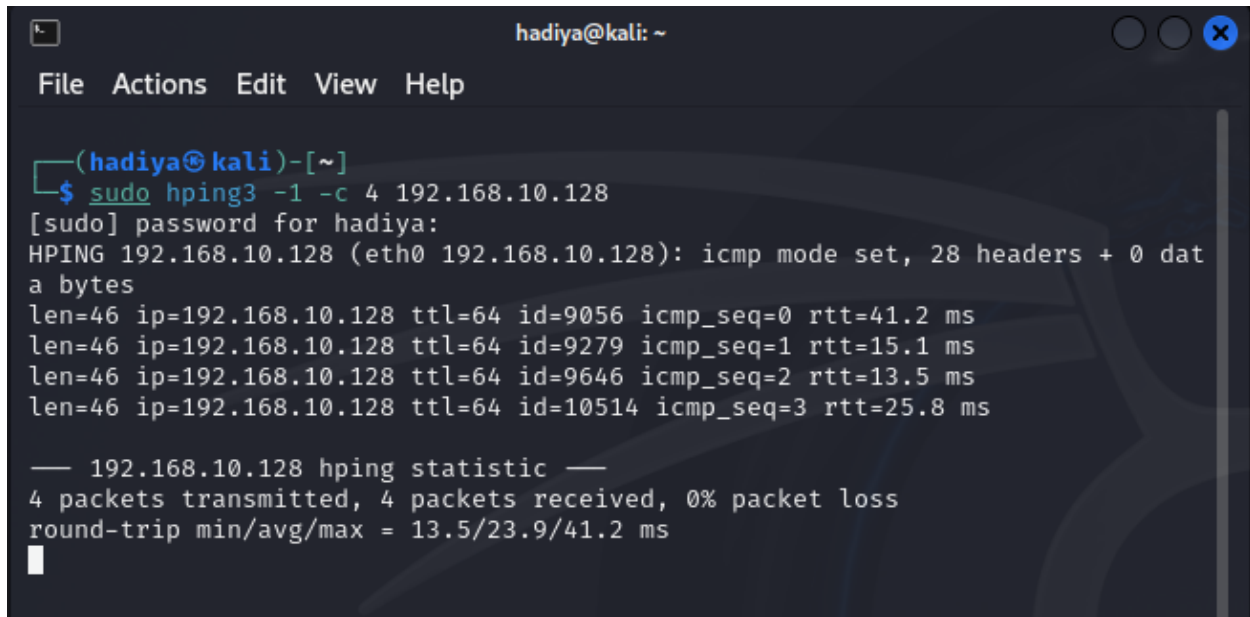
`sudo apt install hping3 -y`

```
hadiya@kali: ~  
File Actions Edit View Help  
(hadiya@kali)-[~]  
$ sudo apt update  
[sudo] password for hadiya:  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Ign:1 http://http.kali.org/kali kali-rolling InRelease  
Err:1 http://http.kali.org/kali kali-rolling InRelease  
Temporary failure resolving 'http.kali.org'  
All packages are up to date.  
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease  
Temporary failure resolving 'http.kali.org'  
Warning: Some index files failed to download. They have been ignored, or old  
ones used instead.  
Notice: Repository 'Kali Linux' changed its 'firmware component' value from '  
non-free' to 'non-free-firmware'  
Notice: More information about this can be found online at: https://www.kali.  
org/blog/non-free-firmware-transition/  
(hadiya@kali)-[~]  
$ sudo apt install hping3 -y  
hping3 is already the newest version (3.a2.ds2-11~kali1).  
hping3 set to manually installed.  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
(hadiya@kali)-[~]  
$
```

Task 01 Basic Ping request:

1. Manually send ICMP echo requests to ubuntu machine from Kali:

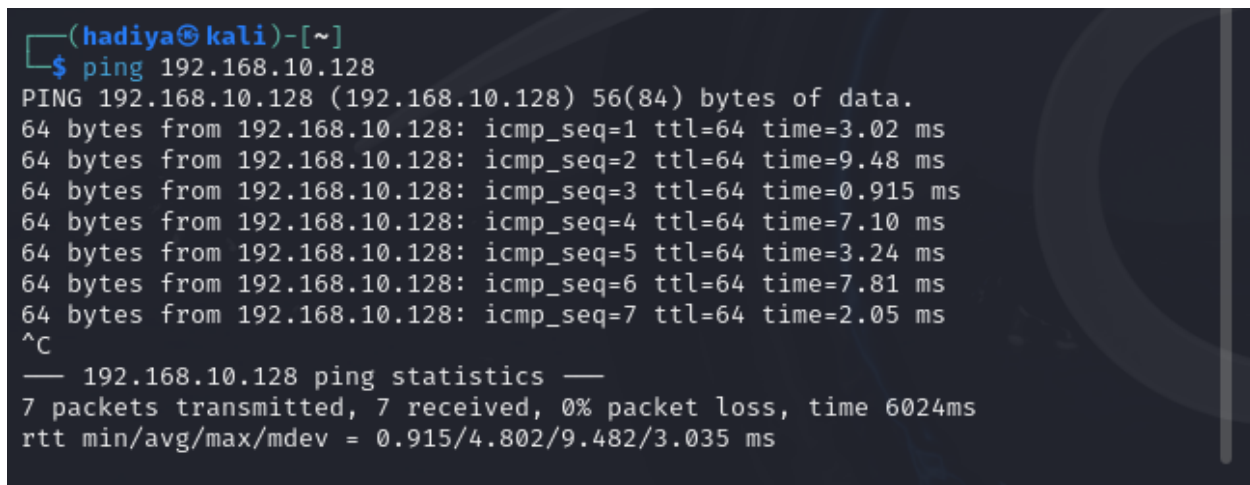
Sudo hping3 -I -c 4 target-ip

A terminal window titled 'hadiya@kali: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(hadiya@kali)-[~]'. The command '\$ sudo hping3 -I -c 4 192.168.10.128' is entered. The output shows the password for 'hadiya' and then the results of the hping3 command: 'HPING 192.168.10.128 (eth0 192.168.10.128): icmp mode set, 28 headers + 0 data bytes'. It lists four packets with their lengths, IP addresses, TTLs, IDs, ICMP sequence numbers, and round-trip times (rtt). A summary line shows '192.168.10.128 hping statistic' with '4 packets transmitted, 4 packets received, 0% packet loss' and 'round-trip min/avg/max = 13.5/23.9/41.2 ms'.

```
(hadiya@kali)-[~]
$ sudo hping3 -I -c 4 192.168.10.128
[sudo] password for hadiya:
HPING 192.168.10.128 (eth0 192.168.10.128): icmp mode set, 28 headers + 0 data bytes
len=46 ip=192.168.10.128 ttl=64 id=9056 icmp_seq=0 rtt=41.2 ms
len=46 ip=192.168.10.128 ttl=64 id=9279 icmp_seq=1 rtt=15.1 ms
len=46 ip=192.168.10.128 ttl=64 id=9646 icmp_seq=2 rtt=13.5 ms
len=46 ip=192.168.10.128 ttl=64 id=10514 icmp_seq=3 rtt=25.8 ms

— 192.168.10.128 hping statistic —
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 13.5/23.9/41.2 ms
```

2. On Kali now simply use *ping* (target ip) for normal system ping

A terminal window titled 'hadiya@kali: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(hadiya@kali)-[~]'. The command '\$ ping 192.168.10.128' is entered. The output shows 'PING 192.168.10.128 (192.168.10.128) 56(84) bytes of data.' followed by seven lines of ping results, each showing '64 bytes from 192.168.10.128: icmp_seq=X ttl=64 time=X.XX ms'. A summary line shows '192.168.10.128 ping statistics' with '7 packets transmitted, 7 received, 0% packet loss, time 6024ms' and 'rtt min/avg/max/mdev = 0.915/4.802/9.482/3.035 ms'.

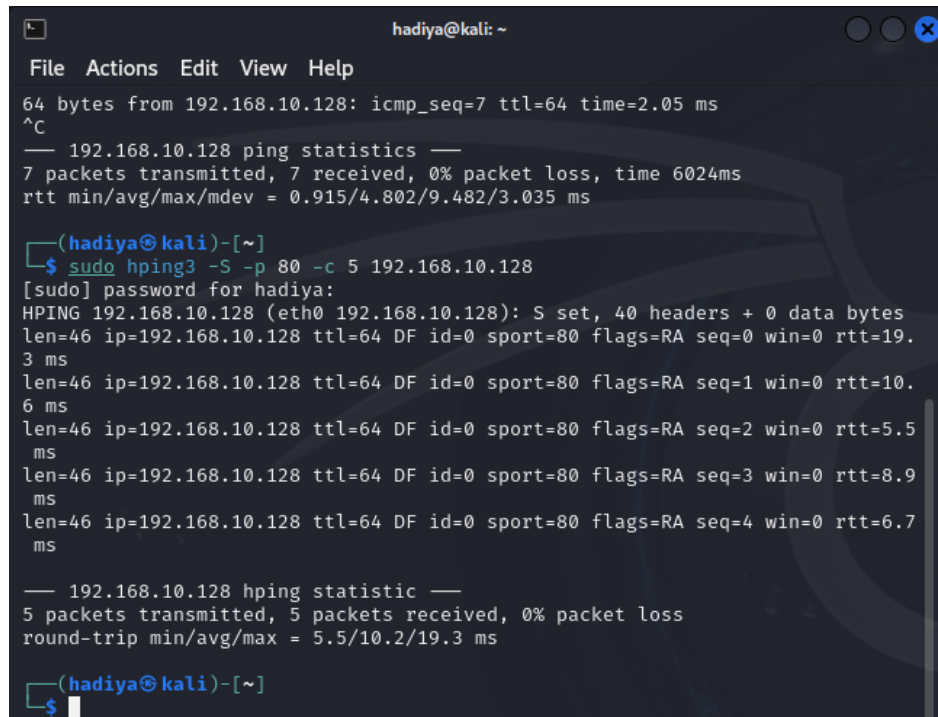
```
(hadiya@kali)-[~]
$ ping 192.168.10.128
PING 192.168.10.128 (192.168.10.128) 56(84) bytes of data.
64 bytes from 192.168.10.128: icmp_seq=1 ttl=64 time=3.02 ms
64 bytes from 192.168.10.128: icmp_seq=2 ttl=64 time=9.48 ms
64 bytes from 192.168.10.128: icmp_seq=3 ttl=64 time=0.915 ms
64 bytes from 192.168.10.128: icmp_seq=4 ttl=64 time=7.10 ms
64 bytes from 192.168.10.128: icmp_seq=5 ttl=64 time=3.24 ms
64 bytes from 192.168.10.128: icmp_seq=6 ttl=64 time=7.81 ms
64 bytes from 192.168.10.128: icmp_seq=7 ttl=64 time=2.05 ms
^C
— 192.168.10.128 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 6024ms
rtt min/avg/max/mdev = 0.915/4.802/9.482/3.035 ms
```

We now know that with hping3 we have more control whereas ping demonstrates OS default behaviour

Task 02 TCP port Scanning:

- If port is open:
Target replies with SYN-ACK.
- If port is closed:
Target replies with RST.

sudo hping3 -S -p 80 -c 5 <192.168.10.128>



```
hadiya@kali: ~
File Actions Edit View Help
64 bytes from 192.168.10.128: icmp_seq=7 ttl=64 time=2.05 ms
^C
— 192.168.10.128 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 6024ms
rtt min/avg/max/mdev = 0.915/4.802/9.482/3.035 ms

(hadiya@kali)-[~]
$ sudo hping3 -S -p 80 -c 5 192.168.10.128
[sudo] password for hadiya:
HPING 192.168.10.128 (eth0 192.168.10.128): S set, 40 headers + 0 data bytes
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=0 win=0 rtt=19.3 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=1 win=0 rtt=10.6 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=2 win=0 rtt=5.5 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=3 win=0 rtt=8.9 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=4 win=0 rtt=6.7 ms
— 192.168.10.128 hping statistic —
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 5.5/10.2/19.3 ms

(hadiya@kali)-[~]
$
```

Kali machine

3. For scanning a range of ports:

- Sending SYN packets to multiple ports.
- Checking which ports respond differently.

This is reconnaissance phase in real attacks.

sudo hping3 -S -p 20-100 <192.168.10.128>

```
(hadiya@kali)-[~]
$ sudo hping3 -S -p 20-100 192.168.10.128
HPING 192.168.10.128 (eth0 192.168.10.128): S set, 40 headers + 0 data bytes
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=20 flags=RA seq=0 win=0 rtt=12.7 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=20 flags=RA seq=1 win=0 rtt=9.2 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=20 flags=RA seq=2 win=0 rtt=8.5 ms
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=20 flags=RA seq=3 win=0 rtt=3.0 ms
```

PART 2: DoS SIMULATION

STEPS:

Task 01 Run a SYN flood Attack:

- Sending thousands of SYN packets per second.
- Not completing handshake.
- Filling target's connection queue.

The server:

- Allocates memory for each half-open connection.
- Eventually resources get exhausted.

This is a **SYN Flood attack**.

Attack target is TCP connection table.

```
sudo hping3 -S --flood -p 80 <192.168.10.128>
```

```
(hadiya@kali)-[~]
$ sudo hping3 -S --flood -p 80 192.168.10.128
HPING 192.168.10.128 (eth0 192.168.10.128): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.10.128 hping statistic —
52183 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(hadiya@kali)-[~]
$
```

Task 02 ICMP flood Attack:

- Flooding ICMP Echo Requests.
- Target must respond to each request.
- Consumes CPU and bandwidth.

This is **ICMP flood attack**.

Attack target:

Bandwidth & processing power.

sudo hping3 -1 --flood <192.168.10.128>

```
(hadiya@kali)-[~]
$ sudo hping3 -1 --flood 192.168.10.128
[sudo] password for hadiya:
HPING 192.168.10.128 (eth0 192.168.10.128): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.10.128 hping statistic —
50078 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
(hadiya@kali)-[~]
$
```

Task 03 UDP flood Attack:

- Sending large volume of UDP packets.
- No handshake required.
- Target processes each packet blindly.

UDP is dangerous because:

- No connection setup.
- Harder to track state.

This is **UDP flood attack**.

sudo hping3 --udp --flood -p 53 <target-ip>

```
hadiya@kali: ~  
File Actions Edit View Help  
  
(hadiya@kali)-[~]  
$ sudo hping3 --udp --flood -p 53 192.168.10.128  
[sudo] password for hadiya:  
HPING 192.168.10.128 (eth0 192.168.10.128): udp mode set, 28 headers + 0 data  
bytes  
hping in flood mode, no replies will be shown  
^C  
— 192.168.10.128 hping statistic —  
38708 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms  
  
(hadiya@kali)-[~]  
$
```

Task 04 FIN Scan:

Normally FIN is used to:
Close a TCP connection.

But here:

You send FIN without connection existing.

If port open:

No response.

If closed:

RST response.

This is called a stealth scan.

Stealth because no full handshake occurs.

sudo hping3 -F -p 80 <target-ip>

```
(hadiya@kali)-[~]  
$ sudo hping3 -F -p 80 192.168.10.128  
HPING 192.168.10.128 (eth0 192.168.10.128): F set, 40 headers + 0 data bytes  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=0 win=0 rtt=6.2  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=1 win=0 rtt=6.0  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=2 win=0 rtt=5.6  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=3 win=0 rtt=8.7  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=4 win=0 rtt=8.2  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=5 win=0 rtt=3.1  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=6 win=0 rtt=6.2  
ms  
len=46 ip=192.168.10.128 ttl=64 DF id=0 sport=80 flags=RA seq=7 win=0 rtt=5.1  
ms
```


PART 3: Wireshark Analysis

STEPS:

Open wireshark by typing “wireshark” in CMD

Click the Host- Only interface connected to Kali machine. Start capturing.

Task 01 Scan:

1. Run SYN flood attack using above command and observe

Wireshark interface showing a packet capture on the ens33 interface. The packet list displays a series of SYN packets from 192.168.10.129 to 192.168.10.128. The packet details show the TCP header with Seq=0, Win=512, Len=0. The packet bytes show the raw data.

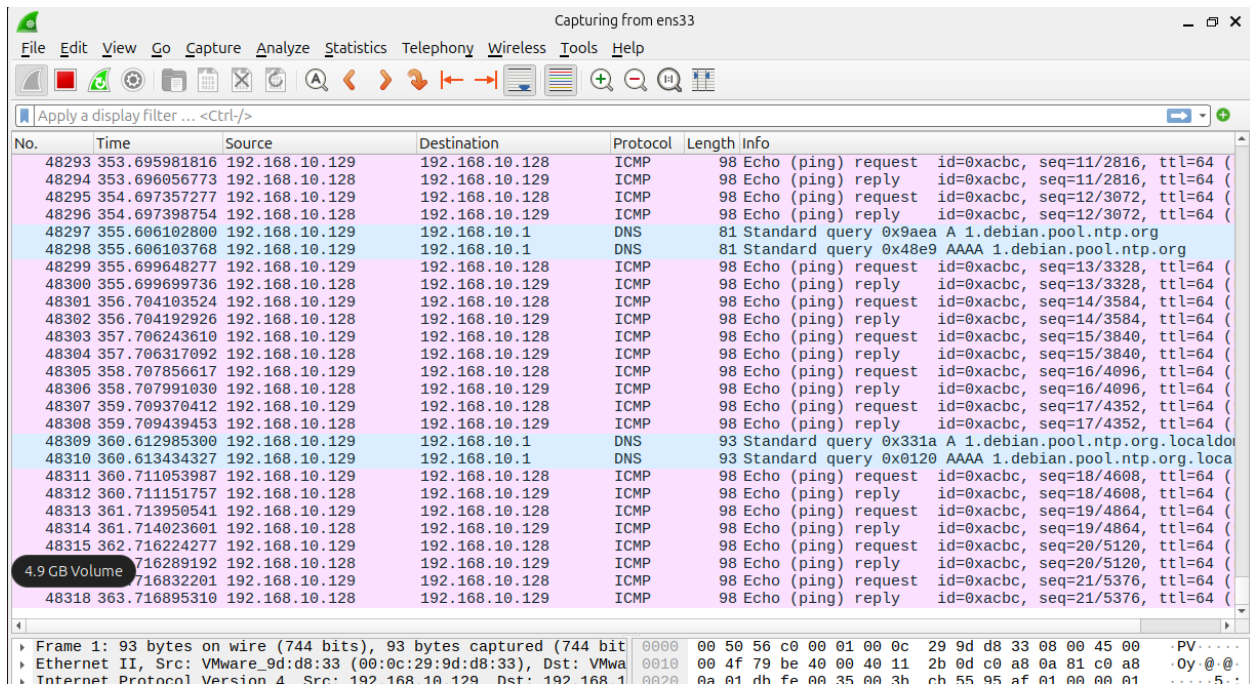
No.	Time	Source	Destination	Protocol	Length	Info
47604	207.465743498	192.168.10.129	192.168.10.128	TCP	60	25841 → 80 [SYN] Seq=0 Win=512 Len=0
47605	207.465765630	192.168.10.128	192.168.10.129	TCP	54	80 → 25841 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47606	207.466316111	192.168.10.129	192.168.10.128	TCP	60	25842 → 80 [SYN] Seq=0 Win=512 Len=0
47607	207.466338292	192.168.10.128	192.168.10.129	TCP	54	80 → 25842 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47608	207.467089943	192.168.10.129	192.168.10.128	TCP	60	25843 → 80 [SYN] Seq=0 Win=512 Len=0
47609	207.467119251	192.168.10.128	192.168.10.129	TCP	54	80 → 25843 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47610	207.468762943	192.168.10.129	192.168.10.128	TCP	60	25844 → 80 [SYN] Seq=0 Win=512 Len=0
47611	207.468763172	192.168.10.129	192.168.10.128	TCP	60	25845 → 80 [SYN] Seq=0 Win=512 Len=0
47612	207.468787822	192.168.10.128	192.168.10.129	TCP	54	80 → 25844 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47613	207.468910570	192.168.10.128	192.168.10.129	TCP	54	80 → 25845 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47614	207.469408357	192.168.10.129	192.168.10.128	TCP	60	25846 → 80 [SYN] Seq=0 Win=512 Len=0
47615	207.469429529	192.168.10.128	192.168.10.129	TCP	54	80 → 25846 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47616	207.470054316	192.168.10.129	192.168.10.128	TCP	60	25847 → 80 [SYN] Seq=0 Win=512 Len=0
47617	207.470075712	192.168.10.128	192.168.10.129	TCP	54	80 → 25847 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47618	207.470581030	192.168.10.129	192.168.10.128	TCP	60	25848 → 80 [SYN] Seq=0 Win=512 Len=0
47619	207.470609975	192.168.10.128	192.168.10.129	TCP	54	80 → 25848 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47620	207.471043955	192.168.10.129	192.168.10.128	TCP	60	25849 → 80 [SYN] Seq=0 Win=512 Len=0
47621	207.472104395	192.168.10.129	192.168.10.128	TCP	60	25850 → 80 [SYN] Seq=0 Win=512 Len=0
47622	207.472130460	192.168.10.128	192.168.10.129	TCP	54	80 → 25849 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47623	207.472233992	192.168.10.128	192.168.10.129	TCP	54	80 → 25850 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47624	207.473757908	192.168.10.129	192.168.10.128	TCP	60	25851 → 80 [SYN] Seq=0 Win=512 Len=0
47625	207.473758215	192.168.10.129	192.168.10.128	TCP	60	25852 → 80 [SYN] Seq=0 Win=512 Len=0
47626	207.473780328	192.168.10.128	192.168.10.129	TCP	54	80 → 25851 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47627	207.473926078	192.168.10.128	192.168.10.129	TCP	54	80 → 25852 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47628	207.474558075	192.168.10.129	192.168.10.128	TCP	60	25853 → 80 [SYN] Seq=0 Win=512 Len=0
47629	207.474578975	192.168.10.129	192.168.10.128	TCP	54	80 → 25853 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47630	207.475118203	192.168.10.129	192.168.10.128	TCP	60	25854 → 80 [SYN] Seq=0 Win=512 Len=0

Microsoft Search (Alt+Q)

Just start typing here to bring features to your fingertips and get help.

Frame 1: 93 bytes on wire (744 bits) captured (744 bits) on interface ens33, Dst: VMwa...
Ethernet II, Src: VMwa...
Internet Protocol Version 4, Src: 192.168.10.129, Dst: 192.168.10.128

2. Run normal ping from kali



Capturing from ens33

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
48293	353.695981816	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=11/2816, ttl=64 (
48294	353.696056773	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=11/2816, ttl=64 (
48295	354.697357277	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=12/3072, ttl=64 (
48296	354.697398754	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=12/3072, ttl=64 (
48297	355.606102800	192.168.10.129	192.168.10.1	DNS	81	Standard query 0x9aea A 1.debian.pool.ntp.org
48298	355.606103768	192.168.10.129	192.168.10.1	DNS	81	Standard query 0x48e9 AAAA 1.debian.pool.ntp.org
48299	355.699648277	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=13/3328, ttl=64 (
48300	355.699699736	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=13/3328, ttl=64 (
48301	356.704103524	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=14/3584, ttl=64 (
48302	356.704192926	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=14/3584, ttl=64 (
48303	357.706243610	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=15/3840, ttl=64 (
48304	357.706317092	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=15/3840, ttl=64 (
48305	358.707856617	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=16/4096, ttl=64 (
48306	358.707991030	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=16/4096, ttl=64 (
48307	359.709370412	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=17/4352, ttl=64 (
48308	359.709439453	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=17/4352, ttl=64 (
48309	360.612985300	192.168.10.129	192.168.10.1	DNS	93	Standard query 0x331a A 1.debian.pool.ntp.org.localdo
48310	360.613434327	192.168.10.129	192.168.10.1	DNS	93	Standard query 0x0120 AAAA 1.debian.pool.ntp.org.loca
48311	360.711053987	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=18/4608, ttl=64 (
48312	360.711151757	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=18/4608, ttl=64 (
48313	361.713950541	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=19/4864, ttl=64 (
48314	361.714023601	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=19/4864, ttl=64 (
48315	362.716224277	192.168.10.129	192.168.10.128	ICMP	98	Echo (ping) request id=0xacbc, seq=20/5120, ttl=64 (
	716289192	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=20/5120, ttl=64 (
	716832201	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) request id=0xacbc, seq=21/5376, ttl=64 (
48318	363.716895310	192.168.10.128	192.168.10.129	ICMP	98	Echo (ping) reply id=0xacbc, seq=21/5376, ttl=64 (

4.9 GB Volume

Frame 1: 93 bytes on wire (744 bits), 93 bytes captured (744 bit)
Ethernet II, Src: VMware_9d:d8:33 (00:0c:29:9d:d8:33), Dst: VMwa
Internet Protocol Version 4, Src: 192.168.10.129, Dst: 192.168.1

3. Stop capture

Task 02 Feature Extraction from Normal & Malicious Traffic

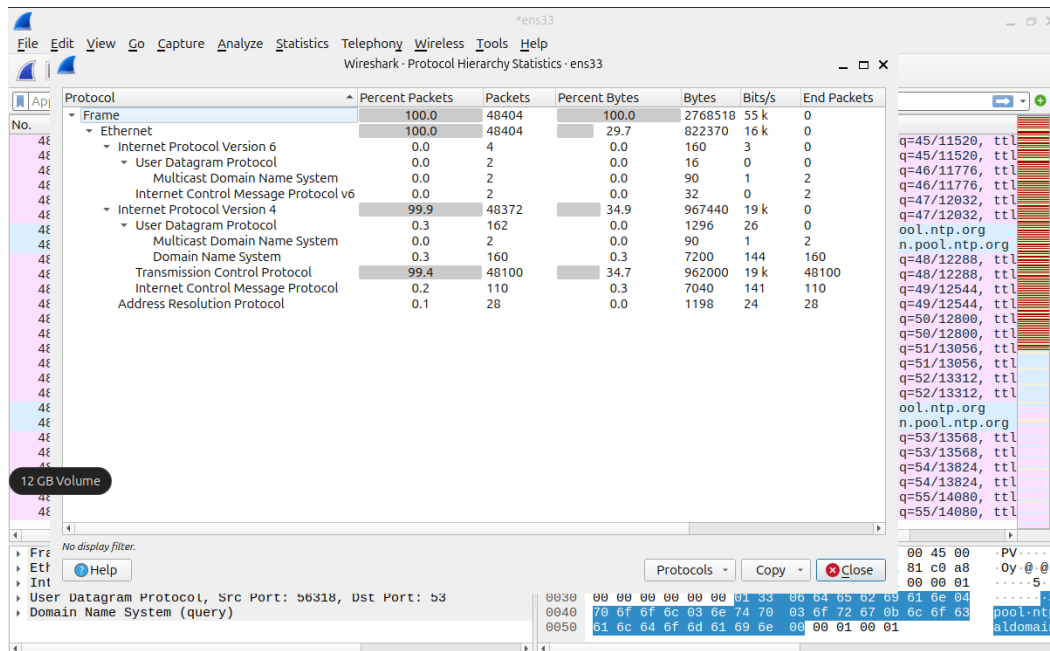
➤ Extract features such as:

- Packet Size
- Inter-packet Arrival Time
- Source/Destination IP & Port Analysis
- Protocol Distribution
- Flow Duration

Identify Protocols:

1. Go to:

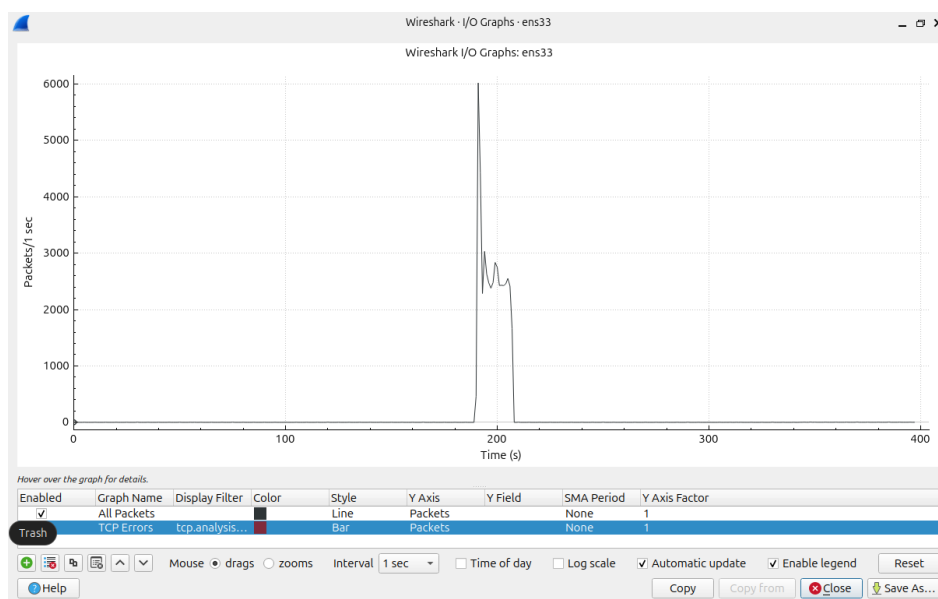
Statistics → Protocol Hierarchy



During SYN flood simulation, TCP packets constituted the majority of traffic, indicating abnormal protocol concentration compared to normal traffic conditions.

IO Graph:

2. Go to: **Statistics → IO Graph**

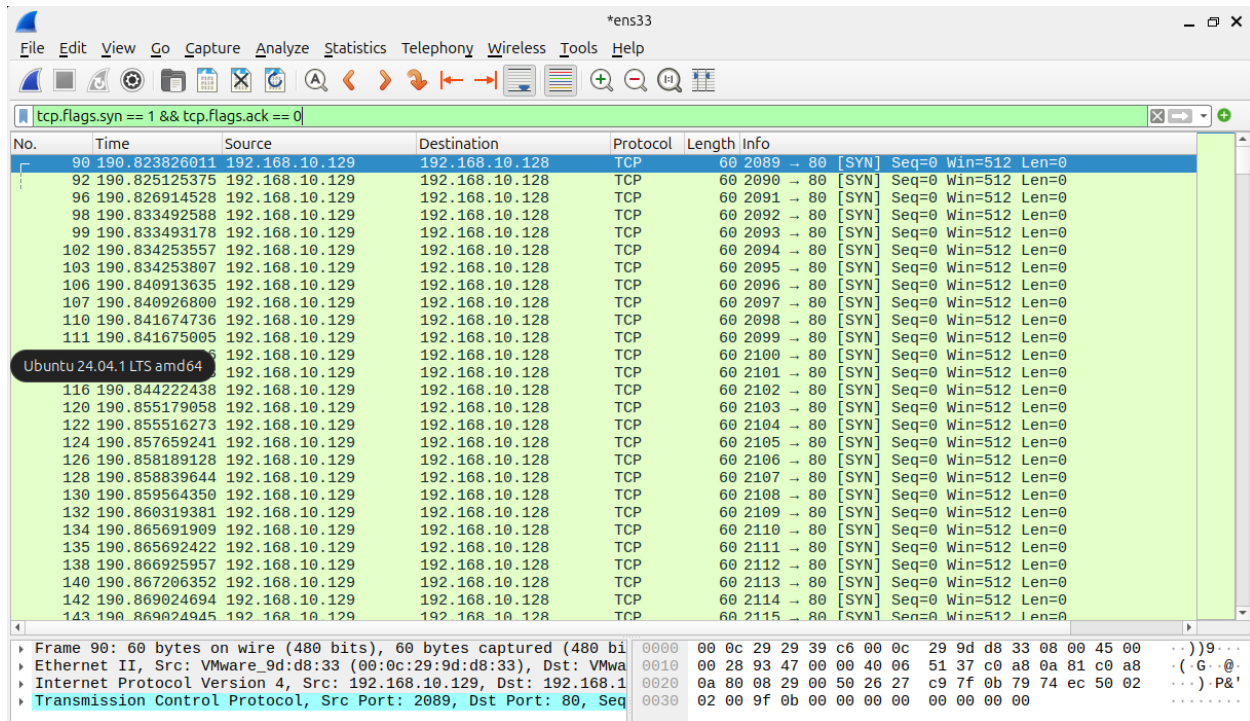


The IO graph displayed a sudden spike in packet rate during the attack interval, clearly distinguishing malicious traffic from normal network behavior.

Analyze SYN Flood specifically

3. Apply filter:

tcp.flags.syn == 1 && tcp.flags.ack == 0



No.	Time	Source	Destination	Protocol	Length	Info
90	190.823826011	192.168.10.129	192.168.10.128	TCP	60	2089 → 80 [SYN] Seq=0 Win=512 Len=0
92	190.825125375	192.168.10.129	192.168.10.128	TCP	60	2090 → 80 [SYN] Seq=0 Win=512 Len=0
96	190.826914528	192.168.10.129	192.168.10.128	TCP	60	2091 → 80 [SYN] Seq=0 Win=512 Len=0
98	190.833492588	192.168.10.129	192.168.10.128	TCP	60	2092 → 80 [SYN] Seq=0 Win=512 Len=0
99	190.833493178	192.168.10.129	192.168.10.128	TCP	60	2093 → 80 [SYN] Seq=0 Win=512 Len=0
102	190.834253557	192.168.10.129	192.168.10.128	TCP	60	2094 → 80 [SYN] Seq=0 Win=512 Len=0
103	190.834253807	192.168.10.129	192.168.10.128	TCP	60	2095 → 80 [SYN] Seq=0 Win=512 Len=0
106	190.840913635	192.168.10.129	192.168.10.128	TCP	60	2096 → 80 [SYN] Seq=0 Win=512 Len=0
107	190.840926800	192.168.10.129	192.168.10.128	TCP	60	2097 → 80 [SYN] Seq=0 Win=512 Len=0
110	190.841674736	192.168.10.129	192.168.10.128	TCP	60	2098 → 80 [SYN] Seq=0 Win=512 Len=0
111	190.841675005	192.168.10.129	192.168.10.128	TCP	60	2099 → 80 [SYN] Seq=0 Win=512 Len=0
		Ubuntu 24.04.1 LTS amd64	192.168.10.128	TCP	60	2100 → 80 [SYN] Seq=0 Win=512 Len=0
			192.168.10.128	TCP	60	2101 → 80 [SYN] Seq=0 Win=512 Len=0
116	190.844222438	192.168.10.129	192.168.10.128	TCP	60	2102 → 80 [SYN] Seq=0 Win=512 Len=0
120	190.855179058	192.168.10.129	192.168.10.128	TCP	60	2103 → 80 [SYN] Seq=0 Win=512 Len=0
122	190.855516273	192.168.10.129	192.168.10.128	TCP	60	2104 → 80 [SYN] Seq=0 Win=512 Len=0
124	190.857659241	192.168.10.129	192.168.10.128	TCP	60	2105 → 80 [SYN] Seq=0 Win=512 Len=0
126	190.858189128	192.168.10.129	192.168.10.128	TCP	60	2106 → 80 [SYN] Seq=0 Win=512 Len=0
128	190.858839644	192.168.10.129	192.168.10.128	TCP	60	2107 → 80 [SYN] Seq=0 Win=512 Len=0
130	190.859564350	192.168.10.129	192.168.10.128	TCP	60	2108 → 80 [SYN] Seq=0 Win=512 Len=0
132	190.860319381	192.168.10.129	192.168.10.128	TCP	60	2109 → 80 [SYN] Seq=0 Win=512 Len=0
134	190.865691909	192.168.10.129	192.168.10.128	TCP	60	2110 → 80 [SYN] Seq=0 Win=512 Len=0
135	190.865692422	192.168.10.129	192.168.10.128	TCP	60	2111 → 80 [SYN] Seq=0 Win=512 Len=0
138	190.866925957	192.168.10.129	192.168.10.128	TCP	60	2112 → 80 [SYN] Seq=0 Win=512 Len=0
140	190.867206352	192.168.10.129	192.168.10.128	TCP	60	2113 → 80 [SYN] Seq=0 Win=512 Len=0
142	190.869024694	192.168.10.129	192.168.10.128	TCP	60	2114 → 80 [SYN] Seq=0 Win=512 Len=0
143	190.869024945	192.168.10.129	192.168.10.128	TCP	60	2115 → 80 [SYN] Seq=0 Win=512 Len=0

Frame 90: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: VMware_9d:d8:33 (00:0c:29:9d:d8:33), Dst: VMwa...
Internet Protocol Version 4, Src: 192.168.10.129, Dst: 192.168.1...
Transmission Control Protocol, Src Port: 2089, Dst Port: 80, Seq...

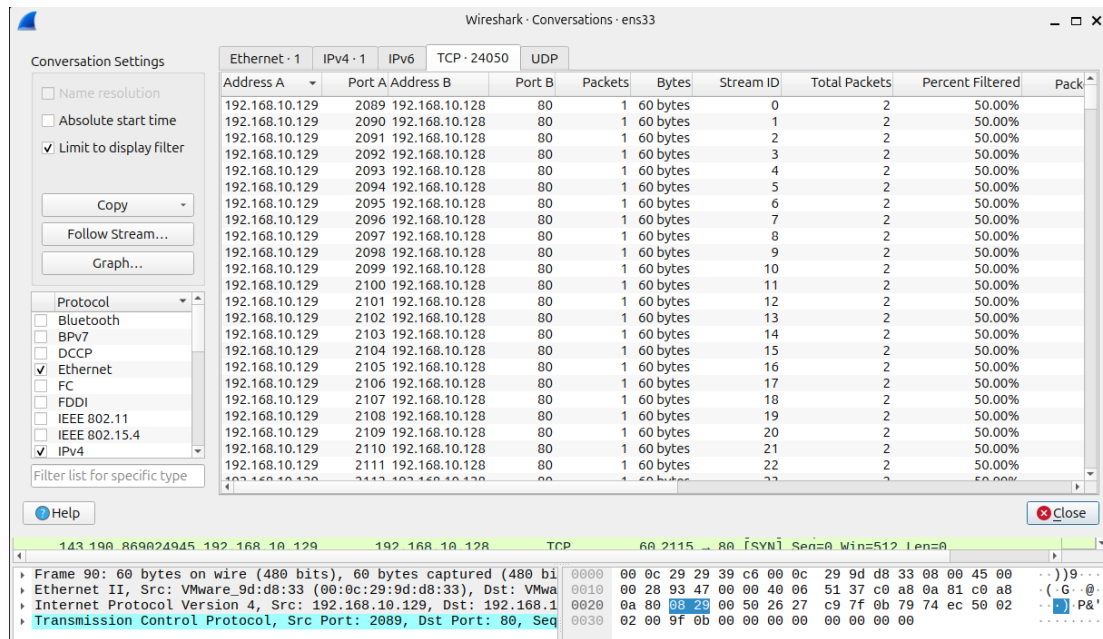
A high volume of SYN packets without corresponding ACK responses was observed, indicating incomplete TCP handshakes characteristic of a SYN flood attack.

Conversations Analysis

4. Go to:

Statistics → Conversations

Click TCP tab.



The screenshot shows the Wireshark 'Conversations' window for the 'ens33' interface. The 'TCP' tab is selected. The table lists 22 connections, all originating from 192.168.10.129 and targeting 192.168.10.128 on port 80. Each connection consists of a single packet of 60 bytes. The 'Percent Filtered' column shows 50.00% for all entries. The 'Conversation Settings' panel on the left shows 'Limit to display filter' checked. The bottom pane shows the details of the selected packet (Frame 90), identifying it as a TCP SYN packet from 192.168.10.129 to 192.168.10.128 on port 80.

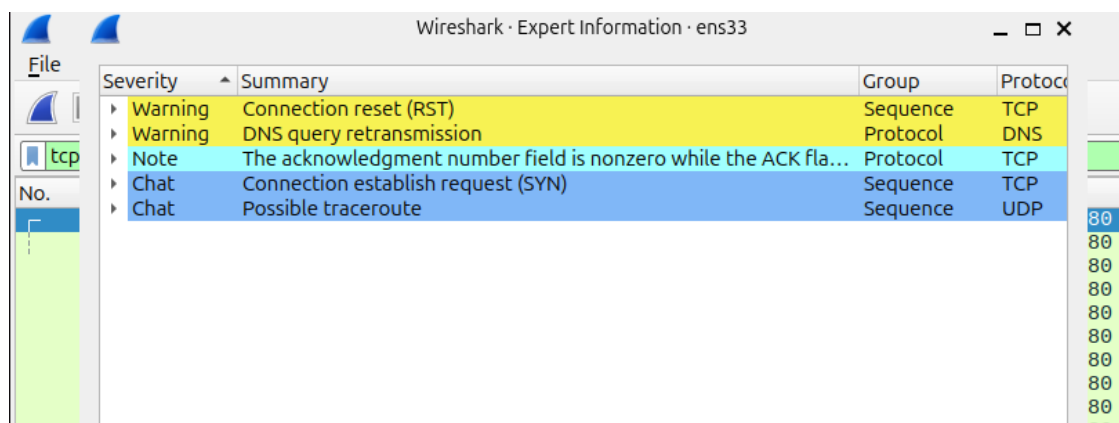
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered
192.168.10.129	2089	192.168.10.128	80	1	60 bytes	0	2	50.00%
192.168.10.129	2090	192.168.10.128	80	1	60 bytes	1	2	50.00%
192.168.10.129	2091	192.168.10.128	80	1	60 bytes	2	2	50.00%
192.168.10.129	2092	192.168.10.128	80	1	60 bytes	3	2	50.00%
192.168.10.129	2093	192.168.10.128	80	1	60 bytes	4	2	50.00%
192.168.10.129	2094	192.168.10.128	80	1	60 bytes	5	2	50.00%
192.168.10.129	2095	192.168.10.128	80	1	60 bytes	6	2	50.00%
192.168.10.129	2096	192.168.10.128	80	1	60 bytes	7	2	50.00%
192.168.10.129	2097	192.168.10.128	80	1	60 bytes	8	2	50.00%
192.168.10.129	2098	192.168.10.128	80	1	60 bytes	9	2	50.00%
192.168.10.129	2099	192.168.10.128	80	1	60 bytes	10	2	50.00%
192.168.10.129	2100	192.168.10.128	80	1	60 bytes	11	2	50.00%
192.168.10.129	2101	192.168.10.128	80	1	60 bytes	12	2	50.00%
192.168.10.129	2102	192.168.10.128	80	1	60 bytes	13	2	50.00%
192.168.10.129	2103	192.168.10.128	80	1	60 bytes	14	2	50.00%
192.168.10.129	2104	192.168.10.128	80	1	60 bytes	15	2	50.00%
192.168.10.129	2105	192.168.10.128	80	1	60 bytes	16	2	50.00%
192.168.10.129	2106	192.168.10.128	80	1	60 bytes	17	2	50.00%
192.168.10.129	2107	192.168.10.128	80	1	60 bytes	18	2	50.00%
192.168.10.129	2108	192.168.10.128	80	1	60 bytes	19	2	50.00%
192.168.10.129	2109	192.168.10.128	80	1	60 bytes	20	2	50.00%
192.168.10.129	2110	192.168.10.128	80	1	60 bytes	21	2	50.00%
192.168.10.129	2111	192.168.10.128	80	1	60 bytes	22	2	50.00%

The Conversations analysis revealed a significant concentration of packets originating from a single source IP targeting one destination, indicating a DoS pattern.

Expert Info

5. Go to:

Analyze → Expert Information



The screenshot shows the Wireshark 'Expert Information' window. It displays a list of events with columns for Severity, Summary, Group, and Protocol. The events include a Warning for a connection reset (RST), a Warning for a DNS query retransmission, a Note about a nonzero acknowledgment number field, and two Chat messages for a connection establishment request (SYN) and a possible traceroute.

Severity	Summary	Group	Protocol
Warning	Connection reset (RST)	Sequence	TCP
Warning	DNS query retransmission	Protocol	DNS
Note	The acknowledgment number field is nonzero while the ACK fla...	Protocol	TCP
Chat	Connection establish request (SYN)	Sequence	TCP
Chat	Possible traceroute	Sequence	UDP

Expert Information highlighted retransmissions and duplicate ACKs during the attack phase, suggesting network stress and abnormal traffic conditions.

Expert Info – Detected events and anomalies

Severity	Summary	Protocol	Interpretation
Warning	Connection reset (RST)	TCP	TCP connection was abruptly terminated.
Warning	DNS query retransmission	DNS	DNS request was retransmitted because no response was received in time.
Note	The acknowledgement number field is nonzero	TCP	TCP packets should have the ACK flag set if the acknowledgment number is nonzero. This note suggests a protocol anomaly
Chat	Connection establish request (SYN)	TCP	This represents the initial step in the TCP three-way handshake to establish a connection.
Char	Possible traceroute	UDP	This packet might be part of a traceroutes operation. Traceroute uses UDP packets (or ICMP) to map the route to a destination.

Malicious Traffic and Normal Traffic

Feature	Normal Traffic	Observed (suspicious/malicious like traffic)
Packet rate	Low and stable	Slight increase due to repeated retransmissions or burst traffic
Source IP Addresses	Few legitimate IP's	Same source IP sending repeated packets
Destination IP	Traffic to different servers	Targeting specific destination IPs
TCP Handshake	Complete handshake	Incomplete or repeated handshake sequences observed
Packet Behaviour	No duplication, normal payloads	Duplicate ACKs, retransmissions, abnormal packet sizes
Flow Duration	Normal session duration	Very short or repeated short sessions
Expert Info Alerts	Minimal Alerts	Wireshark / tcpdump alerts on spurious retransmissions, duplicate ACKs

Conclusion:

In this lab, we configured virtual machines and captured network traffic using tools like **tcpdump** and **Wireshark**. We analyzed packet data to identify normal versus suspicious activity and reinforced our understanding of network protocols and security monitoring. The exercise improved our practical skills in packet analysis and network security fundamentals.